

DOCUMENT 00 · MASTER ENGINEERING BASELINE (MODULE) WITH HOST ANNEXES

MCRS

Modular Captured Ride System · CMFSB
Captured Mechanical Floating Safety Body

Always captured. Soft when you want comfort. Locked when you need control. Settles on passive support when power is gone.

Part A · Master baseline (host-independent module) | Annex E · Electric reference host | Annex H · Hybrid reference host

Field	Controlled value
Document ID	MCRS-DOC-00-R8.02 (Master) · Annex E (R8.02-AE) · Annex H (R8.02-AH)
Issue date	2026-10-09
Founder / Legal Owner	Amirbahador Atai (individual rights holder; assignment to a legal entity is a separate controlled act)
Licensable module	MCRS-MOD-001 (four- or six-tower CMFSB isolation module)
Classification	T1 Confidential / NDA controlled
Document class	Host-neutral Master (module values). Host baselines MCRS-FE-DOC-00-R7.03 (electric) and MCRS-DOC-00-R5.34 (hybrid) are subordinate for module values.
Supersedes	MCRS-DOC-00-R8.01 (all defects from 2026-10-09 correction log closed; host annexes aligned with R5.34 / R7.03)
Status	Proposed baseline — pending owner release and independent technical review. Evidence level L0, TRL 2–3.
Independent Technical Reviewer	To be appointed no later than the earlier of 2026-12-05 and G1 closure
Approval authority	TBD after independent review

This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. This Master is the **controlling host-neutral Document 00 for module values**. Host powertrain numbers appear only in the comparison table (§9.2) or by citation to the host baselines.

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1. Purpose, scope and positioning

This baseline defines the licensable CMFSB isolation module (MCRS-MOD-001) independently of the host powertrain, and two reference hosts that show how the module integrates into a vehicle: an electric host (Annex E) and a hybrid host (Annex H). The module is the product; the hosts are reference applications used for interface, mass, packaging and energy studies.

Part	Role	Lineage
Part A — Master	Module architecture, safety invariants, modes, module parameters, requirements, verification, IP and commercial paths. Contains no host powertrain values except the comparison table in §9.2.	R5.32 / R7.01 / R8.01 module content, harmonised and corrected
Annex E — Electric host	Two battery-electric variants (V-L, V-H): geometry, HV and energy, range screening, mass, loads, EV safety	Aligned with MCRS-FE-DOC-00-R7.03
Annex H — Hybrid host	Two hybrid (ICE + e-drive) variants: geometry, fuel range screening, mass, loads, hybrid safety	Aligned with MCRS-DOC-00-R5.34

One value, one place. Module values live in Part A §7. Host values live in their annex or the dedicated host baselines. A number appears in two places only by citation. Where the two hosts differ, the difference is stated in §9.2 rather than blended. Host powertrain numbers do not appear in Part A except the comparison table.

1.1 Value statement and product framing

Occupant compartments in specialty, defence-utility, ambulance and limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both: the upper body stays mechanically captured at all times, with a COMFORT isolation envelope, a SPORT mechanical lock and passive support on power loss. Reference application RA-1 (ambulance / specialty medical transport) is the context for comfort-related requirements; **no clinical outcome is claimed**.

State	Occupant / operator experience	Architecture behaviour	ID
COMFORT	Road harshness isolated from the cabin	Locks released; macro and micro stages active; leveling active	S2
SPORT	Body feels planted for rapid manoeuvres	All required locks confirmed; isolation stages held	S2S
Power loss	Cabin settles on a known mechanical path	Powered valves fail-closed; passive support; no software required	S4

1.2 What is offered

- **Path A (primary):** licence of the module under a paid Option (SIM-001 run and ICD-001 draft); the buyer may bring its own host.
- **Path B:** host engagement studies using the electric or hybrid reference host; no vehicle sale is implied.
- **Path C:** IP-transaction discussion under counsel (T3 only).

1.3 Revision history

Revision	Scope
R8.00	First module-centric master with Annex E and Annex H
R8.01	Host-neutral wording; Part B restored; hybrid range; Class L sizing; REQ-M16
R8.02 (this issue)	All defects from 2026-10-09 correction log closed: one-value-one-place enforced; document-control statements differentiated; Figure 1 caption corrected; P90 mass RSS vs arithmetic; energy buffer both configs; packaging ≥260/280 mm; lock/S4 logic; IDs uniform MCRS-MOD-001; status tags defined; high-severity independent of RPN; state machine S5–S8 OPEN; IP wording “two-piece vehicle body” / “MCRS”; Annexes aligned with R5.34 / R7.03 (300 kWh electric); ENE-E-001 / ENE-H-001 both named.

2. Executive summary

The CMFSB module places a captured, isolated upper safety body on four or six capture towers. Each tower combines a hydraulic macro stage, a force-bounded Lorentz micro stage, progressive stops and a SPORT positive lock. Safety rests on mechanical capture, passive support and stops — never on the actuator or on software.

Module item	Value	Status
Towers per host	4 (Class S) · 4 or 6 (Class L)	BASELINE
Macro / micro stroke	±50 mm (100 mm total) / ±7.5 mm	TARGET
Force peak / RMS per tower	±2.5 kN / ≈0.7 kN	STUDY
Km	≥63 N/√W at 2–5 mm gap	ASSUMPTION
Core tower mass	P50 est. 48 kg (target ≤55); P90 ≈59 kg RSS / 67 kg arithmetic; limit until G2 = 67 kg	TARGET / OPEN
Corner mass planning	Class S ≈70 kg · M ≈85 kg · L ≈95 kg	STUDY
Control bandwidth / first mode	≥20 Hz / ≥60 Hz	TARGET
Mode time / life	≤3 s / ≥10,000 h equivalent	TARGET
Lateral-g interlock	Inhibit >0.35 g; re-enable <0.30 g	TARGET (temporary)
Module supply	48 V class, galvanically isolated from host HV	STUDY
Evidence level	L0 architecture; TRL 2–3	BASELINE

2.1 Readiness assessment (module level)

Dimension	Position	Principal exposure	Closure vehicle
Architecture	Defined and consistent at L0	No hardware or simulation evidence	SIM-001 (G1), tower bench (G2)
Actuator	Km is an assumption	Thermal and force-gap limits unproven	ACT-001 (G2)
Safety	Invariants and screening FMEA	No HARA / FTA / independence argument	SAF-001 (G2)
Packaging	Module needs ≥260 mm (V-L) / ≥280 mm (V-H) + stop stroke	Neither reference host closes (–370 / –240 mm)	G1 3-D stack, re-baseline
Mass	P50 estimate within target; P90 open	Class S 70 kg corner has no design evidence; V-H six-tower P90 605–815 kg	G1 Class S study, G2 supplier data
IP	Ownership chain and filing OPEN	No filing receipt on record	Counsel, provisional filing
Commercial	Path A Option structure defined	No validated interface set	ICD-001 draft (G1)

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The interface between the two body pieces is the MCRS tower set — the sole mechanical connection in normal operation. Active isolation works inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth vibration isolation within ± 7.5 mm	Force-bounded; never a crash or horizontal path
Macro stage (hydraulic)	Leveling, ± 50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at the seat band	Handling stiffness and load sharing; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into the host platform	Structural continuity

Figure 1 — Two-piece body: the upper cell is captured on MCRS towers above the lower host chassis. V-L: four towers (Class S) · V-H: six towers (Electric) / four towers (Hybrid baseline, six-tower option); three towers shown. Vertical support is passive macro support; horizontal loads go through the capture frame and links; abnormal vertical energy through progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment path
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis structure
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro stage + passive support
Large displacement and leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment structures

The Lorentz stage is an actuator option. If replaced by another actuator family, the safety invariants and interface requirements remain applicable under configuration control.

3.3 Hold function

Hold is an optional, rate-limited secondary restraint (hydraulic orifice and/or friction device at the macro stage). It is distinct from the SPORT lock and from the stops, and is not a primary crash-energy path.

3.4 Qualitative comparison with conventional approaches

Approach	Comfort	Handling stiffness	Positive mechanical lock	Independent capture / crash path	Behaviour on power loss
Soft air / hydro-pneumatic	Good	Reduced in rapid manoeuvres	No	Limited to suspension travel and bump stops	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fixed or fail-soft damping
CMFSB / MCRS (target)	Targeted isolation (unproven at L0)	Targeted via SPORT lock	Yes (SPORT)	Yes — stops, capture frame, links	Settles on passive support, captured

Qualitative only; no measured superiority is claimed at evidence level L0.

4. Safety invariants and safety layers

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV domain galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss (clarified)	Locks remain engaged after power loss only if they were already engaged at the instant of loss (SPORT). From pure COMFORT the locks stay released or engage only at the seat band. Logic frozen in ICD-001 / LCK-001; added to REQ-M17 / REQ-M20.

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement curve (STR-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops and lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent routing
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment is claimed. High-severity ($S \geq 9$) items require explicit pre-G2 review independent of RPN score.

5. Control system, modes and state machine

5.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, vehicle dynamics, high-level requests; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, command bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog and plausibility, where adopted
Corner controllers	Local sensing and actuator current / valve commands, bounded by BCE
BMS	Battery state, HV safety, contactor control, thermal limits
Brake controller	Primary service braking and regenerative coordination

Comfort control may be sophisticated; safety bounds remain simple, deterministic and independently monitorable. The monitor may be called “independent” only after power, clock, processor, software, communication and sensor dependencies are analysed and tested. OTA or calibration changes shall not bypass configuration control.

5.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained (equivalent to S4)
S1	Precharge / self-test	Capture healthy; auxiliary power path checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required positive locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance required
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state → S4 on power loss (engaged locks remain engaged only if already locked; from COMFORT locks stay released). S4 exits only to S1 after power restore and self-test. S2↔S2S only under the lateral-g policy with complete lock confirmation. Full transition table including S5–S8 exit conditions (manual clearance) is required in ICD-001. Detection of power loss ($t_{det,max}$) must be hardware/passive so that “S4 without software” is not contradicted.

5.3 Mode sequences

Transition	Sequence
COMFORT → SPORT	Request accepted (no inhibit, lateral-g met, no fault) → macro drives to seat band → seat contact at required corners → locks engage → all required locks confirmed (incomplete set aborts to COMFORT) → stages held → S2S
SPORT → COMFORT	Allowed only under lateral-g policy → lock release commanded → all report released → macro envelope re-enabled → micro control re-enabled → S2
Power loss from COMFORT	Immediate S4; active force removed; passive support retained; locks stay released
Power loss from SPORT	S4 / fail-locked; locks remain engaged until controlled recovery
Critical fault	S3 / S4 as applicable; no automatic comfort restoration

6. Power-loss behaviour and lateral-g policy

6.1 REQ-M07 power-loss structure

After loss of commanded active power the system shall: (1) detect the loss within $t_{det,max}$ (hardware/passive path); (2) enter S4; (3) close the relevant valves within $t_{valve,max}$; (4) remain inside the capture envelope; (5) stay below z_{max} and v_{max} ; (6) prevent uncontrolled free-float; (7) settle in the defined S4 hydraulic state with passive support at or above the class static floor, with residual capacity and environment stated, under electrical loss and the defined hydraulic failure cases. State definition alone is not acceptance.

Limit to freeze	Where	Verified by
$t_{det,max}$, $t_{valve,max}$	ICD-001 (G2)	Bench + fault injection (L2/L3)
z_{max} , v_{max} , settle limit	ICD-001 (G2)	Single-tower power-cut test
Residual stiffness and damping in S4	ICD-001 (G1/G2)	L2 bench
Static support floor after hose-burst case	STR-001 (G2)	Hydraulic failure test
Energy reserve for safe-state functions	ENE-E-001 / ENE-H-001	Power-loss timing test

6.2 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited immediately when $ a_y > 0.35$ g, for automatic and manual requests
Re-enable	Only after $ a_y $ stays below 0.30 g for the hold time
Driver override	May queue a request; cannot defeat the interlock
Filter and latency	Validated filter and measurement-to-inhibit latency budget, both biased to inhibit
Sensor faults	Validity, plausibility, disagreement thresholds, stale-data handling; a single lateral sensor cannot alone trigger an unsafe transition

Thresholds are temporary TARGET values; SIM-001 and HIL-001 freeze final thresholds, hysteresis, filtering and latency.
PARAM-M21-ROLL-GRAD to be defined at G2.

7. Module mass, hydraulics and parameters

7.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	Target ≤55 kg P50 met by estimate; limit until G2 = 67 kg (conservative)
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share (¼)	8	12	Shared (four-corner reference)
Full corner assembly (Class L)	94	137	STUDY; ≈95 kg in host WBS

Core-tower P90 limit: the previous ≈60 kg limit sat just above RSS (59.4 kg) and accepted independence. Conservative default until G2 supplier correlation data: 67 kg (arithmetic). The ≈60 kg limit is accepted only with correlation evidence.

Decomposition for host WBS (mirrored for Annex E/H): dedicated corner 86/125 kg (P50/P90), shared pump 32/48 kg, distribution 17 kg. These rows are the authoritative source for “See Master §7” references in host mass tables.

7.2 Per-class corner mass and static support floor

Class	Mapped use	Corner P50	Four-corner P50	Static basis per corner	Floor (1.25 × Fmax)
S	Light host (four seats)	≈70 kg (target)	≈280 kg	Occupied share 310–350 kg	≥4.5 kN
M	Intermediate	≈85 kg	≈340 kg	Share 450–550 kg	≥7.0 kN
L	Heavy host	≈95 kg	≈380 kg	Reference max 787.5 kg	≥10.0 kN

Four-corner Class L P90: arithmetic $4 \times 137 = 548$ kg; RSS ≈440 kg. Excess vs 380 kg budget: +60 to +168 kg. Six-tower V-H electric P90: RSS ≈605 kg; arithmetic ≈815 kg. Reserve does not cover fully correlated case.

8. Module electrical and auxiliary energy

Module supply is 48 V class, galvanically isolated from host HV. Peak RMS power for the micro-stage array exceeds the baseline DC-DC rating in both configurations:

- Four-tower: peak $\approx 6.3 \text{ kW} > 3 \text{ kW DC-DC}$ → local energy buffer (48 V battery or supercapacitor) is mandatory.
- Six-tower: peak $\approx 9.45 \text{ kW} > 4 \text{ kW DC-DC}$ → local energy buffer is mandatory.

Therefore a local energy buffer is required for both four-tower and six-tower hosts. Energy reserve for safe-state functions is verified under ENE-E-001 (electric) / ENE-H-001 (hybrid). Power-loss detection and valve closure timing are frozen in ICD-001. The buffer is for comfort performance; the S4 safety case does not rely on it.

8.1 Domain partition

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V class	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock actuation
Host traction HV	High voltage	Isolated from module	e-drive, HV battery (host responsibility)
Local energy buffer	48 V	Module domain	Peak shaving for Lorentz; mandatory both configs

9. Host interface envelope and host comparison

The module interface envelope (mechanical hardpoints, hydraulic ports, electrical connectors, space claim) is defined in ICD-001. Host-specific geometry, mass and energy live in the host baselines and are summarised below for cross-host comparison only.

9.1 Packaging residual (both hosts)

Module alone consumes ≥ 260 mm (V-L: tower 120 + travel 140) and ≥ 280 mm (V-H: tower 140 + travel 140) before stop stroke. Both reference hosts show negative residual at quoted heights (-370 mm V-L, -240 mm V-H). OPEN — P0. FIXED-FOR-ENGAGEMENT geometry is an engagement target, not packaging closure.

9.2 Reference host comparison (citation only)

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV dual e-axle	ICE V8 4.0 L TT + e	BEV 6×6 three e-axes	ICE V12 6.0 L TT + e
Peak power	$\approx 350\text{--}380$ kW	ICE $\approx 430\text{--}450$ + e 150–180	$\approx 480\text{--}550$ kW	ICE $\approx 460\text{--}500$ + e 200–250
On-board energy	300 kWh usable	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh
Range target	≥ 800 km	1,000 km $\leq 90\text{--}95$ km/h	≥ 800 km utility	700–800 km
Towers	4 (Class S)	4 (Class S)	6 (Class L)	4 baseline; 6 option
Curb target	3,350 kg	2,865 kg	5,450 kg	5,150 kg
Design GVW	3,850 kg	3,365 kg	7,000 kg	6,150 kg
Margin to ceiling	150 kg	635 kg	0 kg	850 kg
Packaging residual	-370 mm	-370 mm	-240 mm	-240 mm
Evidence	L0	L0	L0	L0

The electric V-H design GVW equals the 7,000 kg ceiling, so payload is the balancing residual; hybrid hosts carry more margin. V-H tower count differs (six electric, four hybrid baseline with six-tower option); harmonisation is a G1 decision. Detailed host values are controlled by MCRS-FE-DOC-00-R7.03 and MCRS-DOC-00-R5.34.

10. Structural principles

Structural screening loads are based on upper sprung mass, not GVW. Current corner loads assume equal share (± 4 or ± 6) — magnitude screen only. STR-001 must calculate real corner distribution including CG location, occupant/payload placement and load transfer under 1 g braking or cornering.

Progressive stops provide the primary abnormal vertical energy path with a defined force–displacement curve (STR-001). Capture frame and links provide the primary horizontal path. SPORT locks are not a substitute for stops. Stop stroke is not yet in the packaging stack and further reduces residual (OPEN).

11. Requirements (selected module)

ID	Statement (summary)	Status
REQ-M07	Power-loss structure (detection, valves, envelope, settle)	OPEN — ICD-001
REQ-M11	Module interface and ICD compliance	OPEN — ICD-001
REQ-M16	Class L sizing basis and corner mass targets	STUDY
REQ-M17/M20	Lock state on power loss: engage only if already locked; COMFORT stays released	OPEN — LCK-001
REQ-M21	Lateral-g interlock 0.35 / 0.30 g	TARGET (temporary)
REQ-M29	Tower count policy (4 vs 6); host-specific decision at G1	OPEN — G1

12. Safety analysis

Screening FMEA exists at L0. No HARA/FTA or independence argument yet. Closure vehicle: SAF-001 (G2). High-severity ($S \geq 9$) items (loss of capture, link failure) require explicit pre-G2 review independent of RPN. RPN scoring does not hide high-severity rows.

ID	Failure mode	S	Action
F-01	Loss of geometric capture	10	Pre-G2 review; redundant paths
F-02	Link / carriage fracture	9	Pre-G2; STR-001
F-04	Power-loss valve fails open	9	Fail-closed design; test
F-05	HV fault propagates to module	9	Galvanic isolation; HVIL

13. Verification

Evidence ladder L0–L5. Current position L0 (architecture). Gate sequence G0–G5. SIM-001 (G1) and tower bench (G2) are the first hardware/simulation vehicles. All transitions of the state machine require HIL evidence before credit.

Level	Meaning	Artefacts
L0	Architecture definition, arithmetic consistency	This baseline
L1	First-order simulation / screening	SIM-001
L2	Component / subsystem bench	Tower bench, ACT-001
L3	HIL / system integration	State-machine HIL, power-loss
L4	Vehicle-level prototype	Instrumented mule
L5	Production-intent / certification	Beyond this baseline

14. Gates, risks and open items

14.1 Open items (module-level selected)

#	Item	Closure
1	Packaging residual both hosts (P0)	G1 3-D datum stack
2	P90 module mass (RSS vs arithmetic); supplier correlation	G2
3	Core tower P90 limit confirmation	G2 supplier data
4	Lock / S4 logic from COMFORT	ICD-001 / LCK-001
5	Full state-machine transition table S5–S8	ICD-001
6	Energy buffer sizing (4- and 6-tower)	ENE-E-001 / ENE-H-001
7	Stop stroke in packaging stack	STR-001 / G1
8	Structural corner load distribution (CG, transfer)	STR-001
9	High-severity (S≥9) pre-G2 review	SAF-001
10	PARAM-M21-ROLL-GRAD definition	G2
11	Km confirmation	ACT-001
12	IP filing receipt	Counsel
13	Independent technical reviewer appointment	≤2026-12-05 or G1
14	V-H tower-count harmonisation (electric 6 / hybrid 4)	G1
15	Class S corner mass design evidence	G1 Class S study

Risk R-22 (packaging P0): residual is certainly negative; probability set to 5 so that $L \times I \geq 20$. Risk R-H2: four-corner P90 mass excess (hybrid lineage).

14.2 Selected risks

ID	Risk	L	I	Score	Mitigation
R-22	Packaging residual remains negative	5	4	20	G1 re-baseline
R-M2	P90 module mass exceeds reserve	4	4	16	G2 supplier correlation
R-M3	Km assumption fails	3	4	12	ACT-001
R-M4	Lock logic ambiguity on power loss	3	4	12	ICD-001 / LCK-001
R-M5	No filing receipt (IP)	3	3	9	Counsel; provisional

15. Controlled document tree

Document / WP	Role	Owner / gate
MCRS-DOC-00-R8.02	Master — module values (this)	Owner; governs module
MCRS-DOC-00-R5.34	Hybrid host baseline	Owner; hybrid values
MCRS-FE-DOC-00-R7.03	Electric host baseline	Owner; electric values
ICD-001	Interface control document	G1 draft
SIM-001	System simulation	G1
ENE-E-001 / ENE-H-001	Energy / range / buffer (electric / hybrid)	G1/G2
SAF-001	Safety analysis / HARA	G2
STR-001	Structural / stops	G2
LCK-001	Lock logic and confirmation	G1/G2
ACT-001	Actuator characterisation	G2
REG-001 / REG-FE-001	Regulatory exposure	G1
THERM-001 / THERM-FE-001	Thermal	G2

16. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai (individual rights holder). Assignment to a legal entity is a separate controlled act. Licensable module: MCRS-MOD-001. Paths A/B/C as defined in §1.2. Wording: integration into a two-piece vehicle body (not “EV body”). Trade-mark reference: MCRS (not “MCRS FULL ELECTRIC”). No filing receipt on record; provisional filing under counsel is OPEN.

17. Claims discipline and glossary

Claim type	Allowed	Forbidden
Range	Host-specific STUDY targets under defined cycle	“Verified range”, clinical outcome
Safety	“Invariants defined; screening FMEA at L0”	“ASIL-rated”, “crash-certified”
Comfort	“Targeted isolation (unproven at L0)”	“Superior ride”
Mass	“P50 closed with reserve; P90 OPEN”	“Mass closed”, “production mass”
Packaging	“FIXED-FOR-ENGAGEMENT / OPEN-P0”	“Locked geometry”, “packaging closed”
Readiness	“TRL 2–3, evidence L0”	“Production ready”, “homologated”

17.1 Glossary (selected)

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body
MCRS	Modular Captured Ride System
MCRS-MOD-001	Licensable four- or six-tower isolation module
COMFORT (S2) / SPORT (S2S) / S4	Isolation mode / locked mode / power-loss capture
Class S / L	Module corner mass / load class
P50 / P90 / RSS	Percentile mass estimates; root-sum-square
FIXED-FOR-ENGAGEMENT / OPEN-P0	Engagement target / blocking packaging item
L0–L5 / G0–G5	Evidence ladder / programme gates
RA-1	Ambulance / specialty medical (comfort context only; no clinical claim)

PART B — Module technical annex

Detailed hydraulic schematics, sensor lists, connector pinouts and CAD envelope live in ICD-001 and supplier-controlled releases. This annex records the governing numerical envelope for the module so that host baselines can cite a single source.

B.1 Stroke and force envelope

Parameter	Value	Status
Macro stroke (body-relative)	±50 mm (100 mm total)	TARGET
Micro stroke	±7.5 mm	TARGET
Force peak per tower	±2.5 kN	STUDY
Force RMS per tower	≈0.7 kN	STUDY
Km	≥63 N/√W at 2–5 mm gap	ASSUMPTION
Control bandwidth	≥20 Hz	TARGET
First structural mode	≥60 Hz	TARGET
Mode transition time	≤3 s	TARGET
Design life	≥10,000 h equivalent	TARGET
Lateral-g inhibit / re-enable	0.35 g / 0.30 g	TARGET temporary
Static support floor Class S / L	≥4.5 kN / ≥10.0 kN	DERIVED

B.2 Hydraulic parameters (screening)

System pressure, orifice sizing, accumulator volume and fail-safe valve specification are frozen in ICD-001 / STR-001. Passive support path (cylinder to accumulator) remains connected on power loss. Hose-burst case static floor is verified under STR-001. Hold function (optional orifice / friction) is rate-limited and is not a primary crash path.

B.3 Sensing and functional independence

Signal	Primary source	Independent / plausibility path
Lateral acceleration	Vehicle dynamics sensor set	Independent channel / cross-check
Yaw and roll rate	Vehicle IMU	Independent channel / model check
Corner position	Dual-channel position sensing where safety-relevant	Cross-corner plausibility
Lock state	Per-lock position / confirmation	Mechanical state correlation
Hydraulic pressure	Pressure transducers	Passive state / rate-of-change plausibility
Battery / HV state	BMS	Independent HV isolation monitor

Redundancy means functional independence, not duplicated identical sensors on one power, processor or signal chain. Common-cause analysis covers shared power, ECU, bus, software, mounting, environment, calibration, vibration and signal processing; a single sensor failure shall not create an unsafe transition.

B.4 Fault classes and recovery

Fault class	Recovery rule
A — transient	S3→S2 only after diagnostic-clear criteria, dwell, sensor plausibility and all interlocks pass
B — latched safety	Authorised reset and recorded review
C — critical capture, structural, lock-integrity or unresolved common-cause	No automatic return to S2 / S2S

B.5 Parameter ID register (module)

PARAM-ID	Description	Value / status	Freeze
PARAM-KM	Lorentz motor constant	$\geq 63 \text{ N}\sqrt{\text{W}}$ at 2–5 mm (ASSUMPTION)	ACT-001 G2
PARAM-REF-MASS-MAX	Reference test mass basis	Host-specific (curb + 200 kg)	SIM-001
PARAM-TOWER-MASS-LIMIT	Core tower P90 limit	67 kg conservative until G2	G2 supplier
PARAM-MODE-TIME	COMFORT $\leftrightarrow$ SPORT transition	$\leq 3 \text{ s}$ TARGET	HIL G3
PARAM-M21-ROLL-GRAD	Lateral-g filter / gradient	TBD	G2
PARAM-FORCE-PEAK	Peak force per tower	$\pm 2.5 \text{ kN}$ STUDY	ACT-001
PARAM-STROKE-MACRO	Macro stroke	$\pm 50 \text{ mm}$ TARGET	ICD-001
PARAM-STROKE-MICRO	Micro stroke	$\pm 7.5 \text{ mm}$ TARGET	ICD-001
PARAM-STATIC-FLOOR-S	Static support floor Class S	$\geq 4.5 \text{ kN}$	STR-001
PARAM-STATIC-FLOOR-L	Static support floor Class L	$\geq 10.0 \text{ kN}$	STR-001

B.6 Space claim and interface summary

Tower structure + mounts: 120 mm (Class S planning) / 140 mm (Class L planning). Macro travel + free play: 140 mm. Module alone therefore claims $\geq 260 \text{ mm}$ (V-L) / $\geq 280 \text{ mm}$ (V-H) before stop stroke. Hardpoint pattern, hydraulic ports, electrical connectors and service access without upper-body removal are defined in ICD-001. The module does not rely on the upper body for primary structural continuity of the lower chassis.

ANNEX E — Electric reference host (aligned with R7.03)

This annex summarises the electric reference host for interface and mass studies. Full control of electric-host values resides in MCRS-FE-DOC-00-R7.03. Module values remain under Part A of this Master.

E1 Variant definition and packaging

Parameter	V-L	V-H
Architecture	BEV 4×4, four towers (Class S)	BEV 6×6, six towers (Class L)
Length / width	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,700 mm	2,000 mm
Wheelbase	3,000 mm	3,300 mm
SPORT ground clearance	200 mm	300 mm
Seats / occupant policy	4 / 400 kg	6 / 600 kg
Battery usable (primary)	300 kWh	300 kWh
Battery gross (≈90 %)	≈333 kWh	≈333 kWh
Curb target	3,350 kg	5,450 kg
Design GVW	3,850 kg	7,000 kg (ceiling; payload residual)
Margin to ceiling	150 kg	0 kg
Packaging residual	−370 mm	−240 mm
Module height claim	≥260 mm before stop	≥280 mm before stop
Optional protection package	—	≈200 kg; identical treatment to hybrid

E2 Powertrain and HV architecture (citation)

Item	V-L	V-H
Nominal HV bus	800 V class	800 V class
Drive topology	Dual e-axle AWD	Front + dual rear e-axes
Peak traction	≈350–380 kW	≈480–550 kW
Continuous traction	≈200–220 kW	≈280–340 kW
AC charger	22 kW base / 44 kW option	22 kW base / 44 kW option
AC 0→100 % time	≈13.6 h / ≈6.8 h	≈13.6 h / ≈6.8 h
DC fast charge	≥150 kW class	≥250 kW class

Pack specific energy 250 Wh/kg ASSUMPTION (aggressive); 180–200 Wh/kg sensitivity reported in R7.03. Design-range capability ≥800 km under defined cycle; "WLTP-class" label removed. Cycle frozen in SIM-FE-001 / ENE-FE-001.

E3 Energy and range screening (300 kWh)

Case	Approx. Wh/km	Range on 300 kWh
V-L ≈100 km/h equivalent	≈240–260	≈1,150–1,250 km
V-L ≈120 km/h	≈300–330	≈910–1,000 km
V-H utility ≈55–60 km/h	≈300–340	≈880–1,000 km
V-H ≈90 km/h	≈400–450	≈670–750 km
V-H ≈120 km/h	≈520–580	≈520–580 km

V-H ≥800 km is tied to low-speed utility cycle. Highway range falls below 800 km.

E4 Mass budget note

V-H P90 MCRS mass with six towers: RSS ≈605 kg; arithmetic ≈815 kg. Reserve (≈220 kg) does not cover the fully correlated case. This elevates risk R-BATT and must be closed at G1/G2 with supplier correlation data. Protection package ≈200 kg treated identically to hybrid for fair comparison. GVW for V-H equals the 7,000 kg ceiling; payload is residual (not an independent input). V-L design GVW 3,850 kg crosses the 3,500 kg threshold in some markets (REG-FE-001).

E5 Module mass reference

See Master §7. Dedicated corner 86/125, shared pump 32/48, distribution 17 (P50/P90). V-H six-tower P50 module ≈565 kg. Core tower P90 limit 67 kg until G2.

E6 Structural screening

Equal-share magnitude screen only. STR-001 for real distribution including CG and load transfer. Upper sprung mass basis, not GVW.

E7 EV safety and host requirements

HVIL, isolation monitoring, precharge, crash isolation, service disconnect, thermal propagation resistance. MCRS safe-state remains available if traction HV is isolated. No ASIL claim. Battery enclosure does not rely on the upper body for primary containment.

E8 Assumptions, risks and open items (electric)

Primary open items: packaging P0, P90 module mass, pack specific energy confirmation, cycle definition for ≥800 km, lock/S4 logic, regulatory 3,500 kg, energy buffer sizing, stop stroke in stack. Full list in R7.03 §17.

ANNEX H — Hybrid reference host (aligned with R5.34)

This annex summarises the hybrid reference host. Full control of hybrid-host values resides in MCRS-DOC-00-R5.34. Module values remain under Part A of this Master.

H1 Host definition and powertrain

Parameter	V-L hybrid	V-H hybrid
Architecture	Hybrid 4×4, four towers (Class S)	Hybrid 6×6, four towers (Class L); six-tower option under G1
Length / width	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,700 mm	2,000 mm
Wheelbase	3,000 mm	3,300 mm
SPORT ground clearance	200 mm	300 mm
Seats	4	6
ICE class	V8 4.0 L twin-turbo class	V12 6.0 L twin-turbo class
ICE power	≈430–450 kW class	≈460–500 kW class
Electric traction / battery	≈150–180 kW / 20 kWh usable	≈200–250 kW / 40 kWh usable
Fuel tank / range target	80 L / 1,000 km (cycle ≤90–95 km/h avg)	150 L / 700–800 km
Curb / design GVW	2,865 / 3,365 kg	5,150 / 6,150 kg
GVW ceiling / margin	4,000 / 635 kg	7,000 / 850 kg
Packaging residual	–370 mm	–240 mm
Optional protection package	—	≈200 kg; identical treatment to electric

V-H six-tower recommendation is justified only on load-per-tower reduction (≈109 → 73 kN at 20 g) and must be evaluated together with the mass penalty (≈185 kg module) and DC-DC ≥4 kW requirement. Decision at G1 (REQ-H06).

H2 Packaging gate

Same stack as electric: residual -370 / -240 mm. Module ≥ 260 mm (V-L) / ≥ 280 mm (V-H) before stop stroke. FIXED-FOR-ENGAGEMENT / OPEN-P0. Illustrative closure levers total 170 mm; remaining residual still negative. Stop stroke and micro-stage clearance further reduce residual (OPEN).

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,700	2,000	FIXED-FOR-ENGAGEMENT
Ground clearance	200	300	FIXED-FOR-ENGAGEMENT
Available for structure	1,500	1,700	DERIVED
Allocated sum	1,870	1,940	DERIVED
Residual margin	-370	-240	OPEN — P0

H3 Fuel range screening (corrected)

η base = 0.22 (band 0.18–0.30) until supplier maps (ENE-H-001). Auxiliary load 1–2 kW included. Prior 0.25–0.30 band was optimistic for twin-turbo V8/V12 at very low load. 1,000 km target (V-L) defined for cycle average ≤ 90 –95 km/h; 90/120 km/h are sensitivity only. V-L does not reach 1,000 km at 120 km/h under any η in the band.

Case	Wheel Wh/km	Range η 0.30 / 0.22 / 0.18
V-L 60 km/h	≈ 115	$\approx 1,860$ / 1,360 / 1,110 km
V-L 90 km/h	≈ 165	$\approx 1,300$ / 940 / 780 km
V-L 120 km/h	≈ 234	≈ 910 / 670 / 550 km
V-H 60 km/h	≈ 244	$\approx 1,640$ / 1,200 / 980 km
V-H 90 km/h	≈ 331	$\approx 1,210$ / 880 / 720 km
V-H 120 km/h	≈ 453	≈ 880 / 650 / 530 km

H4 Controlled mass budget

Four-corner Class L P90: arithmetic 548 kg; RSS ≈ 440 kg. Excess vs 380 kg budget +60 to +168 kg; 150 kg reserve covers only RSS. Risk identifier R-H2 (formerly R-07 in R5.32 lineage). Protection package 200 kg treated identically to electric. Curb V-L 2,865 kg; V-H 5,150 kg (incl. protection). Design GVW 3,365 / 6,150 kg.

H5 Structural screening loads

Equal-share magnitude screen only. STR-001 for real distribution. 20 g four-tower ≈ 109 kN per corner on V-H drives the six-tower option analysis (≈ 73 kN with six towers).

H6 Hybrid safety and host requirements

Fuel, heat and engine excitation near the cell are principal exposures. THERM-001 / SAF-001. HV isolation same invariants as electric. Regulatory note: V-L design GVW 3,365 kg may cross 3,500 kg threshold (REG-001).

H7 Assumptions, risks and open items (hybrid)

Primary open items: packaging P0, V-H tower count (REQ-H06), P90 mass, fuel maps (ENE-H-001), lock/S4 logic, regulatory 3,500 kg, energy buffer, stop stroke. Full list in R5.34 §17.

Appendix A — Correction log summary (R8.01 → R8.02)

Every defect from the 2026-10-09 correction list that applies to the Master or requires Master-level alignment is recorded below.

#	Defect (summary)	Action in R8.02
1–3	P90 mass wording (electric / hybrid)	RSS vs arithmetic ranges stated; reserve coverage explicit
4	Energy buffer only six-tower	Mandatory both 4- and 6-tower
5	Packaging module height	≥260 mm V-L, ≥280 mm V-H
9	Lock vs S4 from COMFORT	Clarified; REQ-M17/M20
10	Document control statements	Master = host-neutral module control; hosts subordinate
11	Figure 1 caption	Six (electric) / four (hybrid baseline, six option)
12	One value one place	Host powertrain only in §9.2 comparison; module by citation
13	ID inconsistency	Uniform MCRS-MOD-001; PARAM table with IDs
14	REQ-M11 / M29	Present in §11
16	Master §7 decomposition	Corner 86/125, pump 32/48, distribution 17 added
17	ENE reference only E	ENE-E-001 / ENE-H-001 both named
18	Risk R-07	R-H2
19–24	Fuel η , range target, pack density, WLTP, tower limit, equal-share loads	Aligned in Annexes E/H and host baselines
28–34	Status tags, R-22, high-S, state machine, regulatory, IP wording	All closed or OPEN with vehicle

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision + owner release
TARGET	Design intent under evidence	G1/G2 evidence can refine
STUDY	Screening only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Hard ceiling / constraint	Programme policy change required

Previous use of “LOCKED” for open packaging heights replaced by FIXED-FOR-ENGAGEMENT throughout.

Appendix C — Work packages, timeline and release

C.1 Primary work packages

WP	Scope	Gate
SIM-001	System simulation, mode transitions, range sensitivity	G1
ICD-001	Mechanical / electrical / hydraulic / software interfaces; lock protocol; power-loss timing	G1 draft / G2 freeze
ENE-E-001 / ENE-H-001	Energy, range, buffer sizing (electric / hybrid)	G1/G2
SAF-001	HARA, FTA, independence argument, high-S review	G2
STR-001	Stops, structural loads, stop stroke	G2
ACT-001	Actuator force–gap–thermal characterisation	G2
LCK-001	Lock logic, confirmation, power-loss behaviour	G1/G2
REG-001 / REG-FE-001	Regulatory exposure including 3,500 kg threshold	G1
THERM-001 / THERM-FE-001	Thermal / exhaust / pack heat rejection	G2

C.2 Work package detail

SIM-001: Multi-body and energy simulation of the two-piece architecture under COMFORT and SPORT, power-loss sequences, and representative road profiles. Outputs: mode transition timing, force/stroke envelopes, range sensitivity. Gate G1.

ICD-001: Mechanical hardpoints, hydraulic ports, electrical connectors, software interfaces, lock confirmation protocol, power-loss timing ($t_{det,max}$, $t_{valve,max}$), lateral-g signal path. Draft at G1; freeze at G2.

ENE-E-001 / ENE-H-001: Supplier pack / fuel maps; buffer energy and recharge rate for both four-tower and six-tower peak RMS; charging curves (electric) or fuel maps (hybrid). Outputs: closed range model, buffer specification.

SAF-001: Full HARA, FTA for capture and power-loss, independence argument for the safety monitor, rescoring of RPN, explicit pre-G2 review of all severity ≥ 9 rows. Outputs: safety case skeleton.

STR-001: Progressive-stop force–displacement, real corner load distribution (CG, payload, transfer), stop stroke for packaging stack, single-tower fail-safe path.

ACT-001: Lorentz (or alternative) force–gap–thermal characterisation, Km confirmation or revision, bandwidth and first-mode measurement.

LCK-001: Positive-lock confirmation protocol, incomplete-set inhibit of SPORT, behaviour on power loss from COMFORT vs SPORT, manual rescue sequence. Shared across hosts.

C.3 Indicative timeline alignment

Ph0a: document release (this Master + host baselines R5.34 / R7.03). Ph0b (months 1–3): SIM-001 start, ICD-001 draft, reviewer appointment, provisional IP filing. G1 (months 3–5): packaging stack, tower-count harmonisation, cycle definition, REG-001. Ph1 (months 5–10): supplier engagement, bench planning, buffer specification. G2 (months 10–14): tower bench, SAF-001, supplier pack / fuel data. The paid Option (SIM-001 + ICD-001 draft) is a bounded subset of Ph0b/G1 deliverables; prior 90-day vs Ph0b inconsistency is closed by this framing. Times are indicative and must be re-baselined with owner and supply-chain constraints.

C.4 Gate exit criteria (summary)

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack; tower-count decision; ICD-001 draft; REG-001; cycle definition
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT/STR/ENE closure
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

C.5 Release control

This document is T1 Confidential / NDA controlled. Distribution limited to parties under valid NDA. Public disclosure of module architecture or claims beyond the claims-discipline table is prohibited until filing and owner release. Independent Technical Reviewer required no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit.

Revision R8.02 supersedes R8.01 for all Master content. Host baselines R5.34 (hybrid) and R7.03 (electric, 300 kWh usable) are subordinate to this Master for all module parameters. Any conflict on module values is resolved in favour of this Master.

Appendix D — Cross-document consistency matrix

This matrix confirms that the three controlling documents (Master R8.02, Hybrid R5.34, Electric R7.03) are aligned on every item that was flagged as inconsistent in the 2026-10-09 correction log.

Item	Master R8.02	Hybrid R5.34	Electric R7.03
Module ID	MCRS-MOD-001	MCRS-MOD-001	MCRS-MOD-001
Document control	Host-neutral module control	Hybrid under Master	Electric subordinate to Master
Figure 1 towers	6 elec / 4 hybrid option	4 baseline / 6 option	6 (Class L)
P90 4-corner	440 RSS / 548 arith	Same	N/A (6-tower)
P90 6-tower	605 RSS / 815 arith	N/A	Same
Energy buffer	Both 4 and 6 mandatory	Same	Same
Module height	≥260 / ≥280 mm	Same	Same
Lock on power loss	Only if already locked	Same	Same
Status tags	Defined Appendix B	Defined §5.1	Defined Appendix B
High S ≥9	Pre-G2 independent of RPN	Same	Same
R-22 probability	5 (≥20)	Same	Same
IP wording	two-piece vehicle body / MCRS	Same	Same
Electric energy	Citation 300 kWh	Citation	300 kWh primary
Fuel η	Citation 0.22 base	0.22 base 0.18–0.30	N/A
Protection 200 kg	Identical both hosts	Same	Same

Appendix E — Full open-item register (module + hosts)

#	Item	Host	Closure
1	Packaging residual –370 / –240 mm (P0)	Both	G1 3-D stack
2	P90 module mass RSS vs arithmetic	Both	G2 supplier
3	Core tower P90 limit 67 kg confirmation	Module	G2
4	V-H tower count harmonisation	Both	G1
5	Lock / S4 logic from COMFORT	Module	ICD/LCK
6	State machine S5–S8 full table	Module	ICD-001
7	Energy buffer sizing both configs	Module	ENE-E/H-001
8	Stop stroke in packaging stack	Both	STR-001 / G1
9	Structural corner distribution	Both	STR-001
10	High-severity pre-G2 review	Module	SAF-001
11	PARAM-M21-ROLL-GRAD	Module	G2
12	Km confirmation	Module	ACT-001
13	Pack specific energy 250 vs 180–200	Electric	Supplier
14	Cycle definition ≥ 800 km	Electric	SIM/ENE
15	Fuel maps η confirmation	Hybrid	ENE-H-001
16	Regulatory 3,500 kg threshold	Both	REG-001
17	IP filing receipt	Module	Counsel
18	Independent technical reviewer	All	$\leq 2026-12-05$ or G1
19	Protection package policy	Both	G1 mass
20	Class S corner mass design evidence	Module	G1 study

Appendix F — Claims discipline (full)

No claim of certified range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate is made in this Master or in the host baselines. Evidence level L0, TRL 2–3. OPEN items remain OPEN until evidence closes them.

Claim type	Allowed statement	Forbidden statement
Range (electric)	"≥800 km design capability under defined cycle (STUDY)"	"800 km verified / certified range"
Range (hybrid)	"1,000 km target for cycle ≤90–95 km/h avg (STUDY)"	"1,000 km verified range at any speed"
Energy	"300 kWh usable TARGET"	"300 kWh certified capacity"
Safety	"Invariants defined; screening FMEA at L0"	"ASIL-rated", "crash-certified"
Comfort	"Targeted isolation (unproven at L0)"	"Superior ride", clinical outcome
Mass	"P50 closed with reserve; P90 OPEN"	"Mass closed", "production mass"
Packaging	"FIXED-FOR-ENGAGEMENT / OPEN-P0"	"Locked geometry", "packaging closed"
Readiness	"TRL 2–3, evidence L0"	"Production ready", "homologated"
Actuator	"Km ≥63 N/√W ASSUMPTION"	"Proven force capability"

Appendix G — Glossary (extended)

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS	Modular Captured Ride System — the product family and licensable module
MCRS-MOD-001	Licensable four- or six-tower isolation module (uniform ID across all documents)
COMFORT (S2)	Isolation mode: locks released, macro + micro active, leveling active
SPORT (S2S)	Locked mode: all required positive locks confirmed, stages held
S4	Power-loss capture: passive support, fail-safe valves, no software dependency for retention
Class S / M / L	Module corner mass / load class (light / intermediate / heavy host)
P50 / P90	50th / 90th percentile mass estimate
RSS	Root-sum-square combination of independent corner uncertainties
FIXED-FOR-ENGAGEMENT	Locked for packaging / engagement studies but not packaging-closed
OPEN-P0	Blocking packaging open item
L0–L5	Evidence ladder levels (architecture → certification)
G0–G5	Programme gates
RA-1	Reference application: ambulance / specialty medical transport (comfort context only; no clinical claim)
BCE	MCRS Body Control ECU
HVIL	High-Voltage Interlock Loop
Usable energy	Net energy available after reserved / inaccessible fraction ($\approx 10\%$)
One value, one place	Module values only in Master; host values only in host baselines or by citation

Appendix H — Document family and supersession

Document	Role	Status
MCRS-DOC-00-R8.02	Master — module values (this)	Current
MCRS-DOC-00-R5.34	Hybrid host baseline	Current; subordinate for module
MCRS-FE-DOC-00-R7.03	Electric host baseline (300 kWh)	Current; subordinate for module
MCRS-DOC-00-R8.01	Prior Master	Superseded by R8.02
MCRS-DOC-00-R5.33	Prior Hybrid	Superseded by R5.34
MCRS-FE-DOC-00-R7.02	Prior Electric	Superseded by R7.03
ICD-001	Interface control	G1 draft pending
SIM-001	System simulation	G1 pending

Any conflict between host baselines on module parameters is resolved in favour of this Master. Host powertrain, energy and mass values are controlled by the respective host baseline.

Appendix I — Safety invariants (restated for standalone reading)

The following invariants are the non-negotiable safety foundation of the module. They appear in Part A §4 and are restated here for readers who use only the annexes.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path.
Power loss (S4)	Passive support; powered valves fail-closed; hydro-pneumatic spring path stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or stops; SPORT never declared with incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat lateral-g interlock or critical fault.
Manual rescue	Two-step rescue under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from module control; single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is lower-chassis responsibility; shall not propagate into upper-body capture.
Single-path rule	No single active-control path credited as primary crash path.
Lock on power loss	Locks remain engaged only if already engaged at loss. From COMFORT locks stay released.

Appendix J — Revision and distribution record

Issue R8.02 dated 2026-10-09. Supersedes R8.01. Distributed under T1 NDA only. Copies outside NDA are uncontrolled and must not be used for design or diligence decisions.

Rev	Date	Author	Summary
R8.00	2026-10-08	Atai	First module-centric master with Annex E/H
R8.01	2026-10-08	Atai	Host-neutral wording; Part B; hybrid range; Class L sizing
R8.02	2026-10-09	Atai	All 2026-10-09 correction-log defects closed; annexes aligned with R5.34 / R7.03

Approval: TBD after independent technical review. This issue is a proposed baseline pending owner release.

Appendix K — Quick-reference module envelope

Single-page numerical envelope for the licensable module MCRS-MOD-001. All values TARGET / STUDY / ASSUMPTION at L0 unless noted.

Parameter	Value
Towers	4 (Class S) or 4/6 (Class L)
Macro stroke	±50 mm (100 mm total)
Micro stroke	±7.5 mm
Force peak / RMS per tower	±2.5 kN / ≈0.7 kN
Km	≥63 N/√W at 2–5 mm (ASSUMPTION)
Core tower P50 / P90	48 kg / 67 kg (limit until G2 = 67 kg)
Corner Class S / L P50	≈70 kg / ≈95 kg
Four-corner P50 / P90	≈280–380 kg / 440–548 kg (RSS–arith)
Six-tower P50 / P90	≈565 kg / 605–815 kg (RSS–arith)
Module height claim	≥260 mm (V-L) / ≥280 mm (V-H) before stop
Control bandwidth / first mode	≥20 Hz / ≥60 Hz
Mode time / life	≤3 s / ≥10,000 h
Lateral-g inhibit / re-enable	0.35 g / 0.30 g (temporary)
Supply	48 V class, galvanically isolated from host HV
Energy buffer	Mandatory for both 4- and 6-tower configurations
Static floor Class S / L	≥4.5 kN / ≥10.0 kN

This envelope is controlled by Part A of this Master. Host baselines cite it; they do not redefine it.

Appendix L — Closing statement

This Master Engineering Baseline R8.02, together with the corrected host baselines MCRS-DOC-00-R5.34 (Hybrid) and MCRS-FE-DOC-00-R7.03 (Electric, 300 kWh usable primary), constitutes the complete set of Document 00 engineering baselines for the MCRS programme as of 2026-10-09.

All numerical, logical, cross-document, assumption and editorial defects identified in the 2026-10-09 correction log have been closed or explicitly carried as OPEN items with named closure vehicles and gates. Module values are controlled exclusively by this Master. Host values are controlled by the respective host baselines. One value, one place.

Evidence level remains L0. TRL remains 2–3. No certification, homologation, ASIL assignment, production readiness, patentability or freedom-to-operate claim is made. Independent technical review is required before any external safety-performance credit.

Legal Owner: Amirbahador Atai. Licensable module: MCRS-MOD-001. Classification: T1 Confidential / NDA controlled. Paths A (module licence Option), B (host engagement studies) and C (IP transaction under counsel) remain as defined.

— END OF MCRS-DOC-00-R8.02 —

Master Engineering Baseline (Corrected) · Module values controlled here · Host baselines subordinate · T1 NDA Controlled · Not Certified